Consider O_2 molecule with electronic configuration: $KK\sigma_{2s}^2\sigma_{2s}^{*2}\sigma_{2pz}^2\pi_{2px}^2\pi_{2py}^2\pi_{2px}^{*1}\pi_{2py}^{*1}$. Calculate the term symbols.

Since, $KK\sigma_{2s}^2\sigma_{2s}^{*2}\sigma_{2pz}^2\sigma_{2px}^2\sigma_{2py}^2$ are completely filled MOs $\rightarrow$ contributions to the total angular momentum component along the molecular axis will be zero.

O_2 molecule has two unpaired electrons.
Here, λ_1 = +1 or -1, λ_2 = +1 or -1
Λ = +2,-2,0

λ_1	+1	-1	λ_1	+1	-1
λ_2	+1	-1	λ_2	-1	+1
Λ	+2	-2	Λ	0	0

Two values of Λ obtained (+2 or -2) are consistent with the fact that $\Lambda \neq 0$, are **doubly degenerate**.

Λ = 0 also correspond to two different ways of arranging λ_i.
In other words, Λ = 0 is also doubly degenerate, <u>which is not possible</u>.

For Λ = +2 or -2, both the electrons are present in the same MO.
Thus, s_1 = +1/2, and s_2 = -1/2, leading to S = 1/2-1/2 = 0, 2S+1 = 2x0+1 = 1

$$\Omega = 2+0, \text{ or } 2-0 = 2$$

The term symbol is $^1\Delta_2$.

For Λ = 0, s_1 = +1/2, s_2 = +1/2 and S = 1, 2S+1 = 3
s_1 = +1/2, s_2 = -1/2 and S = 0, 2S+1 = 1

$$\Omega = 0+1 = 1$$
$$\Omega = 0+0 = 0$$

Term symbols can be $^3\Sigma^-$ and $^1\Sigma^+$.

Since, the **total wave function** ($\Psi_T = \Psi_{space}\Psi_{spin}$) should be <u>anti-symmetric</u>, so anti-parallel spins of electrons has to be combined with Σ^+ (i.e., $^1\Sigma^+$) and parallel spins with Σ^- (i.e., $^3\Sigma^-$).

Further, both the electrons in π^* orbitals have *g*-parity, the total parity will be *g*x*g* = *g*.

The energy level diagrams or spectroscopic states can be based on Hund's rule, i.e., the ground state should have the highest multiplicity.

Further, when two electrons with parallel spin are present in π or δ MOs, it gives rise to Σ^- state.

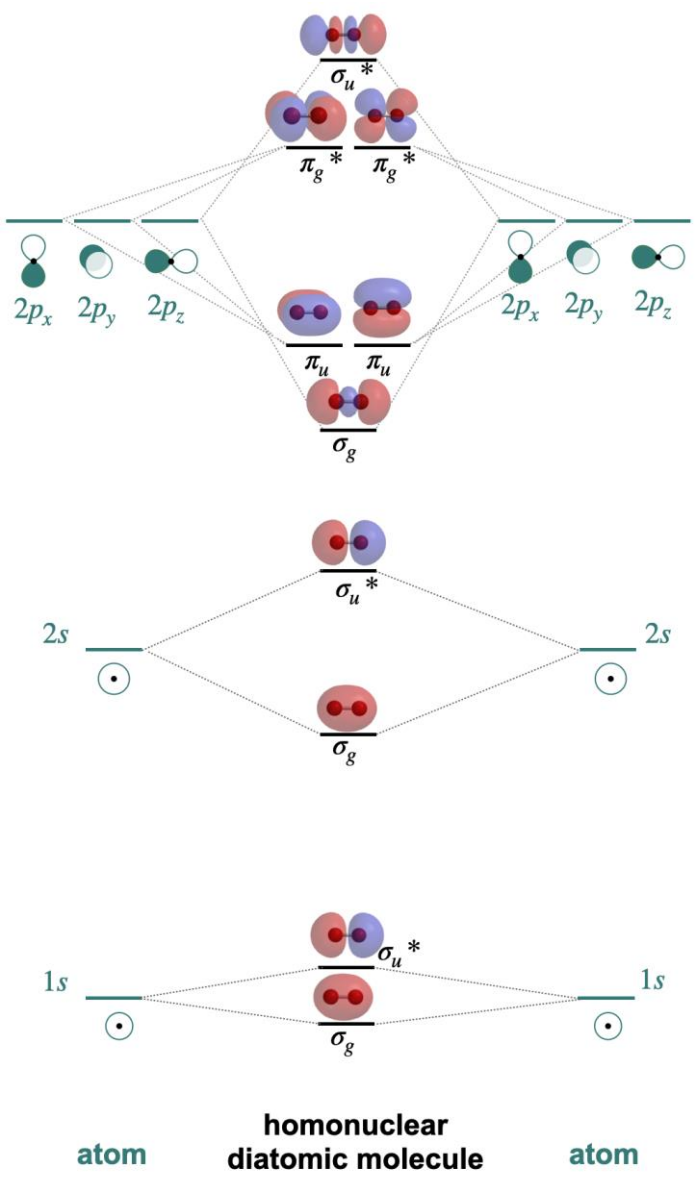
This means that $^3\Sigma_g^-$ is the ground state.

The complete energy level diagram is:

$^1\Sigma g^+$ _____

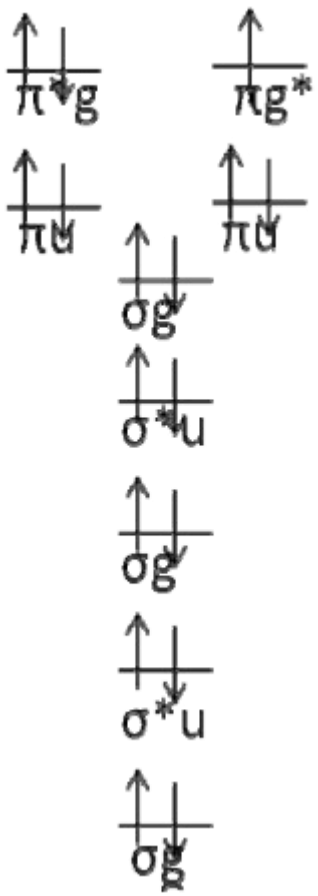
$^1\Delta g$ _____

$^3\Sigma g^-$ _____



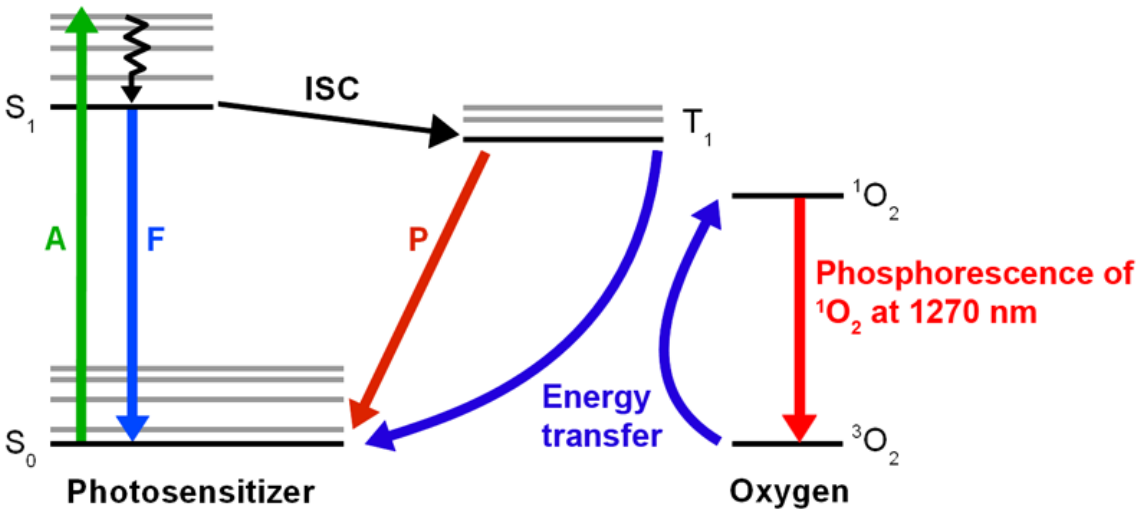
Write the term symbols for O_2^-.

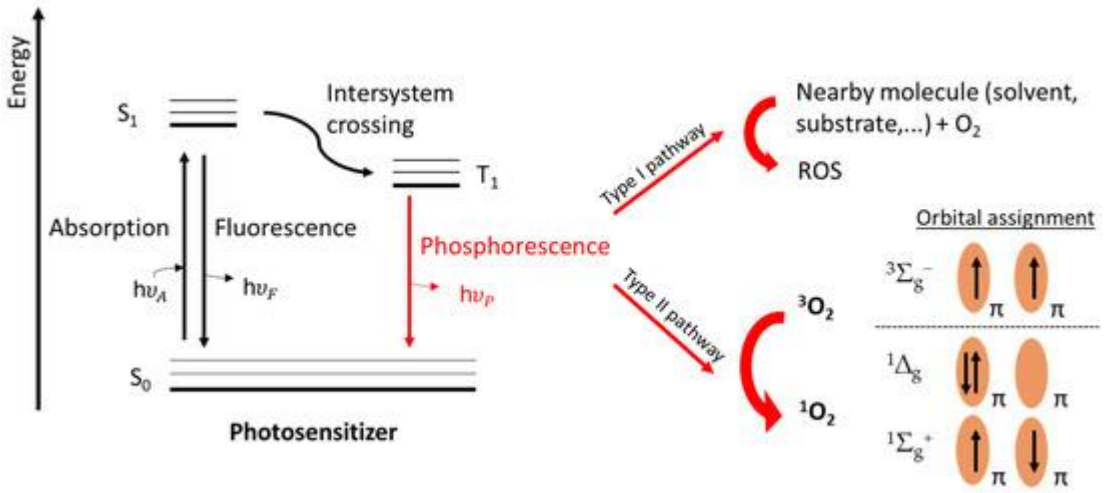
Solution:



There are only two possible configurations. It should be easy to see that the term symbol is $^2\Pi_g$.

Photosensitization and singlet oxygen generation





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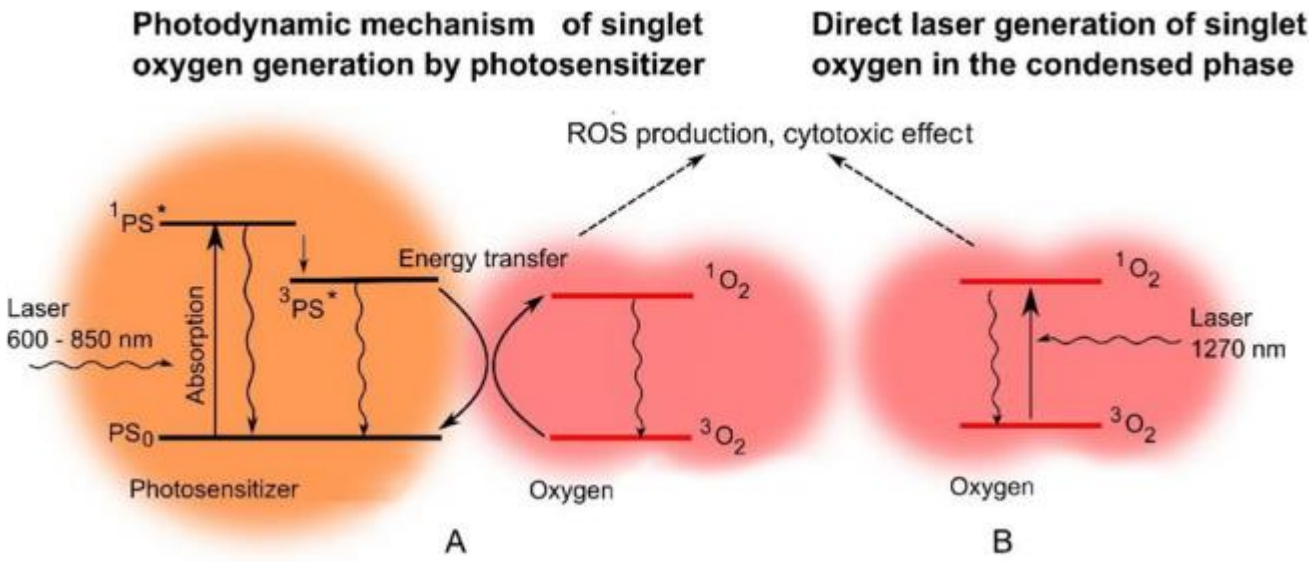
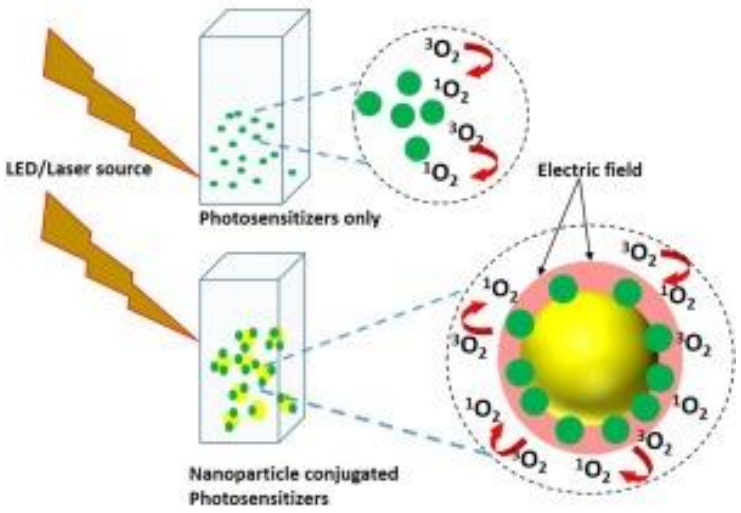




J. Photochem. Photobiol. A, 2017, 332, 66-71

Prog. Quantum Electron., 2017, 55, 112-128

The term symbols for H_2^+ and N_2^+ is $^2\Sigma_g^+$. Explain the meaning of each term.

(i) Σ is the electronic state for which electronic angular momentum quantum number Λ is 0.

(ii) 2 represents the multiplicity of the state, i.e., 2S+1 = 2 for S = +1/2, the resultant electronic spin of the molecule in this state is +1/2.

(iii) The superscript "plus" sign on the right-hand side indicates that the electronic eigen function corresponding to this state does not change sign on reflection at a plane through the internuclear axis.

(iv) The subscript 'g' on the right-hand side represents the parity of the eigen function of the state, which for the given state is even, i.e., the wave function of a molecule does not change sign on reflection through centre of symmetry.

What is the term symbols for N_2? $^1\Sigma_g^+$

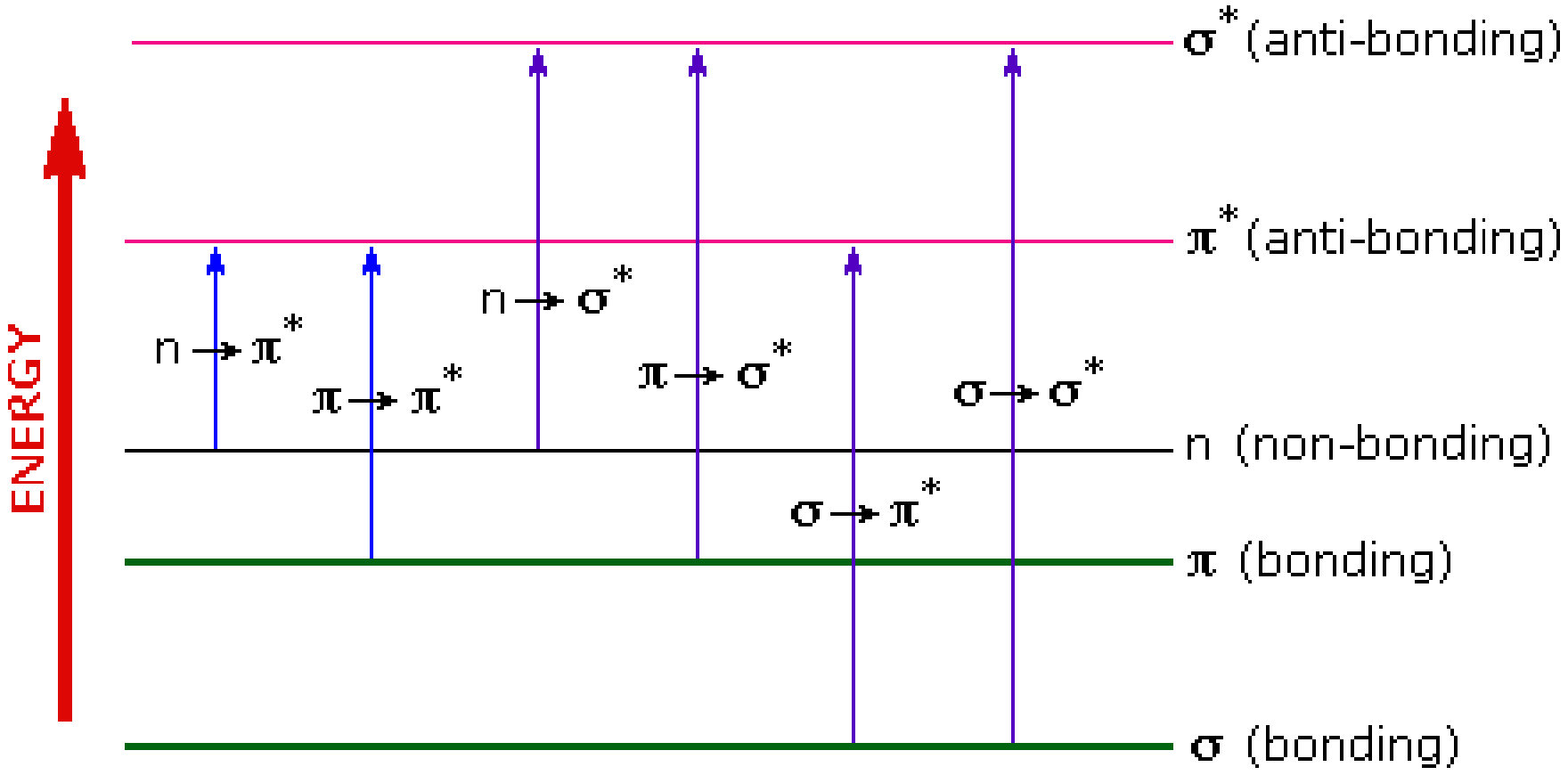
An electron from any of the occupied orbitals can be promoted to higher unoccupied level on absorption of appropriate radiation.

The *n* electrons are most easily excitable and give rise to the longest wave absorption band when excited to π^*-MO $\Rightarrow$ $n \rightarrow \pi^*$ transition.

When an electron from π orbital is promoted

$\Rightarrow$ $\pi \rightarrow \pi^*$ or $\pi \rightarrow \sigma^*$ transition.

$\sigma \rightarrow \sigma^*$ transition are also possible but require much higher energy and may appear in the far-UV.


σ* (anti-bonding)
π* (anti-bonding)
n (non-bonding)
π (bonding)
σ (bonding)
ENERGY
n → π*
π → π*
n → σ*
π → σ*
σ → σ*
σ → π*

On promotion of an electron from any of the occupied orbitals, the energy state of the molecule changes.

These states are designated as (n,π*), (π,π*), (n,σ*), (σ,σ*), (σ,π*), etc. and may have singlet or triplet character.

In the ground state when all the bonding MOs are doubly occupied

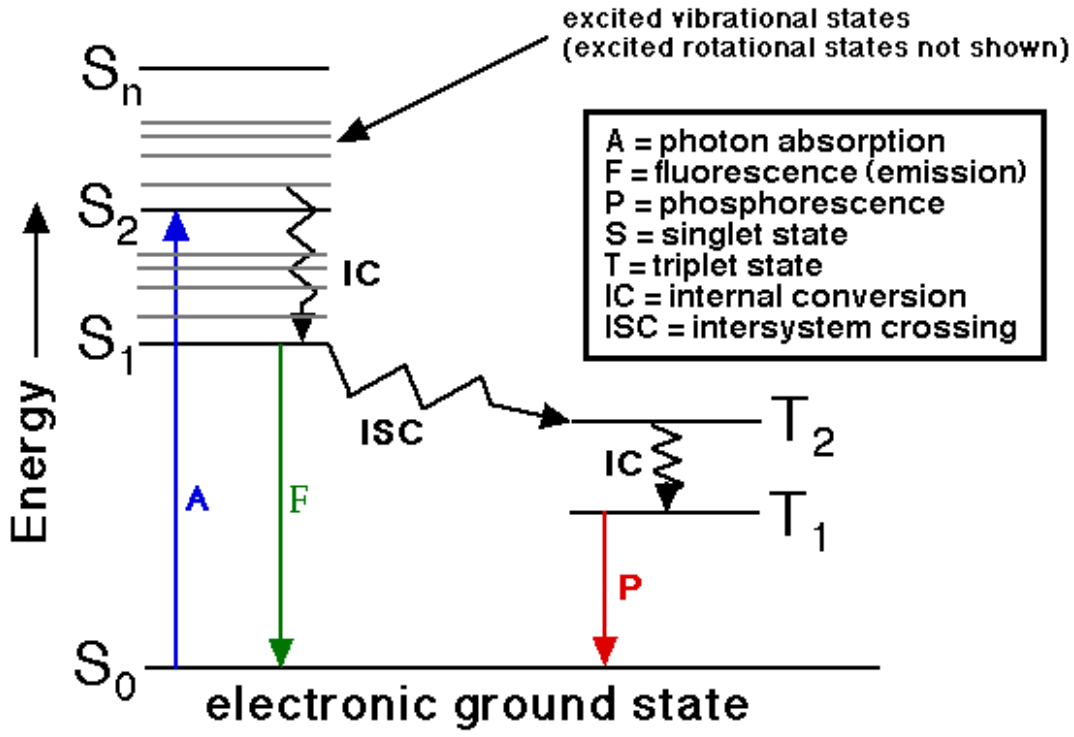
$\rightarrow$ *Pauli exclusion principle* predicts only singlet state.

Once the electrons are orbitally decoupled on excitation, spin restrictions are lifted

$\rightarrow$ *Both S (singlet, spin paired) and T (triplet, spin parallel) states are possible.*

Hund's Rule says the triplet state is the lowest energy excited state.

<u>State diagrams</u> are represented with all the singlet levels expressed as horizontal lines one above the other and the triplet levels are drawn slightly shifted in space, maintaining the order of the energy values.



Such a diagram is known as **Jablonski diagram**.